Lab F1 Is Not a Field Claim: Sealed Packs, Detector Ablation, External OTRF Corpora, and Evidence Boundaries for Multi-Host Correlation

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Artifact: https://github.com/Siddarthb07/corvex

***Abstract*—We study a concrete evaluation-hygiene failure in security tooling: a multi-host campaign correlator can mint sealed held-out micro-F1 1.00 (N_{atk}=6, N_{ben}=6; TP/FP/FN = 6/0/0) while detector-only scores 0.625, then soft-upgrade into field language. We evaluate *Corvex*—HMAC-signed `EventEnvelopes`, weak per-host detectors, optional cross-host fusion, quarantine dry-run only—under five locked protocols with numbers frozen in `NUMBERS_LOCKED.json`. Protocol S (sealed): correlator F1 1.00 vs detector-only 0.625 vs B1 0.00. Protocol B (breaktest/public-TTP): F1 drops to 0.645 vs 0.204 (lift +0.441; TP/FP/FN = 10/10/1 vs 5/33/6). Protocol O (OTRF/Security-Datasets, N=36, external authorship): 34/36 MULTIHOSTNOLIFT, 30× fleet-wide Q dry-run flags—camps form, fusion does not beat detector host coverage. Protocol A (aggressive compounds, N=14): 4 FUSIONHELD (APT29 day1/day2, Caldera APT3, Golden SAML) and 10 SINGLEHOST after named fixes; adapter skip rates on held APT29 remain 0.86–0.94. Protocol L (live laptop replay, N=56): 20 CAUGHT, 19 ALREADYCOVERED, 17 SINGLEHOSTBATCH, 0 MISS; 139,594 events → 55 multi-host camps; mode is offline replay, not malware execution. Lab scores and fusion-lift checks pass; second-host physical telemetry is absent, so we *withhold field claims* and report only the standing synthetic / purple-team scope. We do not claim enterprise validation, SOTA IDS, or live implant catch. Contribution: a falsifiable correlator evaluation with locked protocols that refuse to equate sealed F1 with field readiness.**

***Index Terms*—security evaluation, multi-host correlation, ATT&CK, OTRF, reproducibility, negative results, evaluation hygiene**

1. Introduction: the sealed-F1 failure

Detection and correlation tooling increasingly ships with two surfaces that look similar in a scoreboard and mean radically different things in a paper or deployment brief. The first is a **sealed lab score**: a small held-out synthetic pack that proves the correlator pipeline runs and that fusion beats a detector-only baseline under author-written ground truth. The second is a **field-scope sentence**: an abstract, workshop pitch, or operations brief that says the system "works on multi-host attacks." When those surfaces share a hero F1 and a soft sentence, a perfect sealed pack can mint permanently confident language. The evaluation looks careful. The evaluation world is closed [1]. The reader receives overstated confidence.

This paper documents that failure mode for Corvex, an open-source multi-host campaign correlator [2]. Corvex is the instrument under study; the scientific object is *lab-to-field language drift*: sealed micro-F1 1.00 coexists with breaktest F1 0.645, broad OTRF atomics that never lift host coverage (34/36), and missing second-host evidence that would be required before any enterprise-scope sentence.

Scope contract (not a user study).. If an abstract upgrades to "works on multi-host attacks" whenever sealed F1 is high, then Protocol S's F1 1.00 mints field language while Protocol O shows MULTIHOSTNOLIFT on 34/36 external captures. That is a *measurement-and-scope* failure. We do **not** claim a measured SOC-analyst belief update; we claim that under this common practice, sealed hero numbers are insufficient evidence for field-scope prose (Section 3).

Research questions..

1) Does sealed correlator F1 transfer to harder public-TTP breaktests?
2) On externally authored OTRF captures, does fusion beat detector-only *host coverage*, or only form campaigns?
3) After named correlator/quarantine fixes, which hardest public compounds hold—and what residual breaks remain?
4) Does streaming OTRF into a live correlator process catch multi-host structure under an honest replay mode?
5) Given strong lab scores, what evidence is still missing before a field-scope sentence is warranted?

Answers in brief (locked).. (1) Partial transfer with a hard drop: 1.00 → 0.645 F1; lift remains +0.441 vs detector-only 0.204 (Fig. 1, 2). (2) Camps form; lift fails: 34/36 MULTIHOSTNOLIFT; 30× Q-all-hosts (Fig. 4). (3) 4/14 FUSIONHELD (APT29×2, Caldera APT3, Golden SAML); 10 single-host; APT29 skip_rate 0.86–0.94 remains

(Fig. 5, 6). (4) Live replay N=56: 0 miss; 20 caught; not malware execution (Fig. 7). (5) Lab fusion-lift and integrity checks pass, but second-host physical telemetry is absent—so field-scope claims remain unwarranted (Fig. 8).

Non-claims (stated early).. Not enterprise telemetry validation. Not commercial EDR/XDR parity. Not live malware execution on the author's laptop. Not automatic containment safe to enable (live OS containment remains off). Not a claim that FUSIONHELD on saturated 4–5 host labs equals host-coverage lift over detector-only on every atomic. Anchored claim: evaluation hygiene for multi-host fusion under sealed / breaktest / external / live-replay protocols, with explicit evidence boundaries.

Contributions..

- A five-protocol framing (S/B/O/A/L) plus an evidence checklist (E) that makes lab-to-field drift measurable, with figures rendered from locked JSON.
- Micro-F1 correction: $F1 = 2PR/(P+R)$ from pooled TP/FP/FN; prior macro-of-pack-F1 reporting produced impossible triples.
- External-authorship scoreboards (OTRF N=36, aggressive N=14, live replay N=56) that do not invent Corvex-authored GT F1 on OTRF.
- Named failure mechanisms (ambient metadata poison, recon host-union, fleet-wide quarantine spray, adapter allowlist) with before/after status.
- An explicit withheld field-scope sentence, conditioned on missing second-host evidence rather than on sealed F1 alone.
- Secondary author-fleet batteries reported as engineering probes only (Fleet20; intensive multifactor), never as independence headlines.

Paper roadmap.. Section 3 defines field-scope sentences and evidence boundaries. Section 4 specifies Corvex at the level needed to interpret tables. Section 5 defines protocols S/B/O/A/L/E and metrics. Section 6 reports locked results with figures. Section 7 places author fleets in their proper tier. Section 8 discusses withheld field-scope language and a portable checklist. Sections 9–11 cover related work, threats to validity, and limits.

2. Why this is an evaluation paper

Workshops and journals that care about cybersecurity experimentation ask whether metrics, datasets, and claims are commensurate [3], [4]. Our scoped question is:

> *Can a multi-host correlator report sealed F1* 1.00 *under author ground truth while external purple-team captures and missing second-host evidence still make enterprise-scope prose unwarranted—and can that gap be measured with locked protocols rather than narrative soft upgrades?*

Yes, on the locked Corvex canary. That is an evaluation-hygiene result about one correlator and one protocol ladder, not a theorem about all fusion systems.

Why a small sealed N is still a valid stress.. Critics will say N_{atk}=6 is toy-scale. We agree it is modest. It is, however, exactly the class of score that appears in demos, CI-friendly eval notebooks, and student reimplementations of "campaign correlators." Hero numbers are most tempting precisely in that setting, because the sealed pack is cheap and external corpora are slightly less cheap. If lab-to-field drift appears here, authors of larger stacks should assume they need the same multi-protocol evidence—not that scale magically repairs scope discipline.

3. Threat Model and Evidence Boundaries

3.1. Actors

The **evaluator** wires detectors, fusion, and eval packs. The **reader** (reviewer or practitioner) sees tables and abstracts. The **adversary** need not be sophisticated: distribution shift from author-sealed packs to external purple-team captures is enough. We do not assume an attacker who crafts inputs to evade detectors; we assume a benign author who trusts sealed F1 as field evidence.

Definition 1 (Field-scope sentence)**.** *A field-scope sentence asserts usefulness on real multi-host attacks or enterprise telemetry beyond the authored / purple-team protocols in this paper. Examples: abstract upgrades from "synthetic" to "real," or equating sealed F1 with production readiness.*

Definition 2 (Lab-to-field language drift)**.** *Occurs when (i) a manuscript or brief emits a field-scope sentence keyed to lab scores, (ii) lab micro-F1 is near-perfect on small author-written packs, (iii) harder or external protocols show material drop or no fusion lift, and (iv) the field-scope sentence is not conditioned on those harder protocols.*

Definition 3 (Lab-coherent score)**.** *A score (or set of scores) is lab-coherent when sealed held-out metrics, breaktest fusion lift above a stated minimum, and basic integrity probes are internally consistent under the author's ground truth and tooling. Lab-coherence does* not *imply field-scope warrant.*

3.2. Evidence boundary

We separate lab-coherent scores from field-scope warrant. Second-host physical telemetry is absent in this manuscript, so field-scope sentences remain unwarranted even though lab scores are strong. That separation is the paper's methodological contribution.

3.3. Out of scope

We do not model APT operators adapting to Corvex specifically. We do not claim that fixing language drift fixes detection of novel TTPs. We claim only that score tables and field-scope prose must be tied to named protocols with stated authorship.

4. System Under Test

4.1. Ingest and trust surface

Corvex ingests HMAC-signed `EventEnvelope` records. Enrollment binds producers to hosts; unsigned or mis-enrolled envelopes are rejected. This paper treats ingest integrity as a prerequisite for scoring, not as a novelty claim.

4.2. Detectors (intentionally weak)

Per-host detectors raise weak local hypotheses:

- **Lateral authentication:** unusual auth hops / peer patterns (including WorkstationName-derived peers where available).
- **Micro-exfiltration:** small egress bursts; ambient destinations such as link-local metadata endpoints are poisoned so they cannot freely mint multi-host coverage.
- **Recon fanout:** discovery-style fanout; recon is annotate-oriented and must not union every host into a mega-campaign.
- **Optional DNS beacons:** only when DNS payloads exist (e.g., Sysmon EID 22); otherwise channel health reports a gap.

Detectors are not meant to be individually decisive. The research question for fusion is whether cross-host merge improves campaign recovery relative to grouping alerts by detector key alone.

4.3. Correlator merge

A correlator optionally merges alerts across hosts using time windows, hop graphs, ambient-destination poison, and anti-jumpbox rules (no lateral↔lateral 1-host glue that invents bridges). Transitive merge can still over-merge when shared service accounts or CDN-like egress create spurious bridges; Section 6.6 names those modes.

4.4. Quarantine and sensors

Quarantine modes are `dry_run` / `lab_flag` / `blocked` only. Live OS containment remains off in this evaluation. Surgical dry-run quarantine proposes cut-points rather than spraying every observed host. Stage B Windows/macOS sensors are observe-only. ATT&CK tags are coarse stage hypotheses, not high-confidence attributions.

4.5. What we measure

We measure (i) micro-averaged campaign precision/recall/F1 under author ground truth where it exists, (ii) fusion lift vs detector-only, (iii) observational verdicts on external captures without inventing GT F1, (iv) live-replay catch labels, and (v) whether evidence items required for field-scope prose are present.

TABLE 1. WHAT EACH PROTOCOL SUPPORTS. MIXING SCOPES ACROSS ROWS IS THE CENTRAL HYGIENE FAILURE.

ID	Input	Valid scope
S	Sealed synthetic packs	Wiring + fusion lift under author GT
B	Breaktest public-TTP	Harder author-assembled OOD
O	OTRF atomics $N{=}36$	External authorship; observational only
A	Aggressive compounds $N{=}14$	Hardest public multi-host shape
L	Live replay $N{=}56$	Replay catch labels (not live malware)
E	Evidence checklist	Whether field-scope prose is warranted
F_{sec}	Author fleets (secondary)	Mechanism debugging only

5. Method: Protocols S/B/O/A/L + Evidence Checklist E

5.1. Protocol S — seal held-out packs

Campaign generators write JSONL packs. Packs and scorer rules are Fernet-sealed under an external key before correlator retunes. Sealing reduces accidental train/test snooping; it does *not* create outsider authorship. Current sealed size: attack $N{=}6$ (lateral$\times$2, exfil$\times$2, fusion_chain, recon_lateral) and benign $N{=}6$.

5.2. Protocol B — breaktest

Manifests under `labs/breaktest/` cite public Atomic Red Team / ATT&CK technique IDs [5] but are author-assembled. Results are published even when worse than sealed. Necessary for a non-author fusion-lift check; still not stranger enterprise traffic.

5.3. Protocols O and A — external OTRF

OTRF/Security-Datasets and MITRE ATT&CK Evaluations Caldera/manual captures [6]–[8] are fetched, sanitized (`CommandLine`/`ScriptBlockText` stripped for Defender), adapted, signed under lab enrollment, and scored observationally. Protocol O is the broad atomic set ($N{=}36$). Protocol A is the aggressive compound subset ($N{=}14$: APT29 day1/day2, Caldera/Empire APT3, Golden SAML, LSASS$\times$7, Log4Shell$\times$2).

5.4. Protocol L — live replay

Captures are streamed into a live Corvex process and dash on the author's Windows laptop. Mode name in the lockfile: `offline_lab_replay_into_live_correlator`. This is *not* executing Empire/Covenant/malware on that laptop.

5.5. Evidence checklist E

We record five boolean evidence items (Fig. 8):

1) Non-author fusion lift above a stated minimum (0.05) on breaktest.

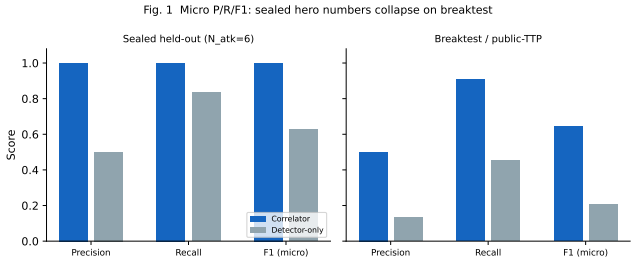


Figure 1. Locked micro P/R/F1. Left: sealed held-out (author GT). Right: breaktest. Correlator vs detector-only. Sealed hero F1 does not survive B.

2) Stranger-operator dry-run attestation (human Ed25519 custody).
3) Sealed benign false-campaign rate at minimum N.
4) Ingest/trust integrity probes (reject bad HMAC; enrollment checks).
5) Elevated telemetry from a *second physical* Windows host.

Lab-coherent items may pass while second-host evidence fails; in that case we withhold field-scope sentences.

5.6. Metrics

Micro-F1 (locked definition).. Let TP/FP/FN be pooled over packs. Then

$$P = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \quad R = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}, \quad F1 = \frac{2PR}{P+R}. \quad (1)$$

Macro mean of per-pack F1 is diagnostic only. A prior draft reported macro-of-F1 alongside micro P/R, yielding impossible triples (e.g., $P \approx 0.33$, $R \approx 0.67$, $F1 \approx 0.67$).

Detector-only ablation.. Same events; alerts grouped by detector key without cross-host merge. Fusion lift $= F1_{\mathrm{corr}} - F1_{\mathrm{det}}$.

Observational OTRF verdicts.. No Corvex-authored ground-truth F1 on OTRF. MULTIHOSTNOLIFT: multi-host campaigns form but correlator does not beat detector-only host coverage. FUSIONHELD: multi-host structure with consolidation and surgical quarantine under the locked scoring rules (on saturated labs this need not imply positive host-coverage lift). SINGLEHOST: capture has < 2 hosts; fusion not testable.

All figures are rendered by `scripts/render_paper_figures.py` from `reports/*.json` with no re-eval at paper build time.

6. Results

6.1. Protocol S — sealed held-out

Benign FCR $= 0.00$ at N_{ben}=6. Perfect correlator F1 is *necessary but not sufficient*. Detector-only 0.625 shows fusion still matters on this set; B1 shows that naive single-host aggregation recovers nothing. Fig. 3 shows per-family sealed correlator F1—still synthetic authorship.

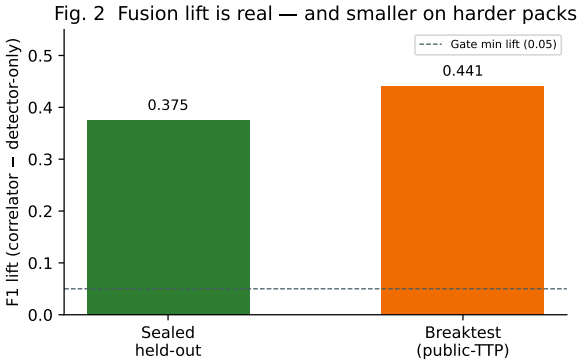


Figure 2. Fusion lift: sealed $+0.375$; breaktest $+0.441$ (dashed line: minimum lift 0.05 used in the non-author check). Lift can grow while absolute F1 falls.

TABLE 2. PROTOCOL S LOCKED COUNTS (AUTHOR-GENERATED, SEALED).

Arm	P	R	$F1$	TP/FP/FN
Correlator	1.00	1.00	**1.00**	6/0/0
Detector-only	0.50	0.833	0.625	5/5/1
B1 single-host	0.00	0.00	0.00	0/1/6

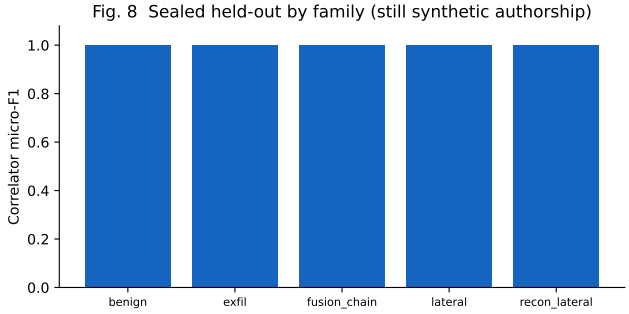


Figure 3. Sealed correlator F1 by family. Still synthetic authorship—do not read as OTRF independence.

TABLE 3. PROTOCOL B LOCKED COUNTS (AUTHOR-ASSEMBLED MANIFESTS; PUBLIC TTP IDS).

Arm	P	R	$F1$	TP/FP/FN
Correlator	0.500	0.909	**0.645**	10/10/1
Detector-only	0.132	0.455	0.204	5/33/6
Lift			$+0.441$	

Interpretation.. Protocol S answers "does the pipeline work under its own sealed packs?" It does not answer "is field-scope prose warranted?"

6.2. Protocol B — breaktest / public-TTP

Absolute F1 drops $1.00 \rightarrow 0.645$. Precision halves ($1.00 \rightarrow 0.50$). False positives rise from 0 to 10 on the correlator arm; detector-only false positives explode to 33. This is the first honest hardness signal: fusion lift remains

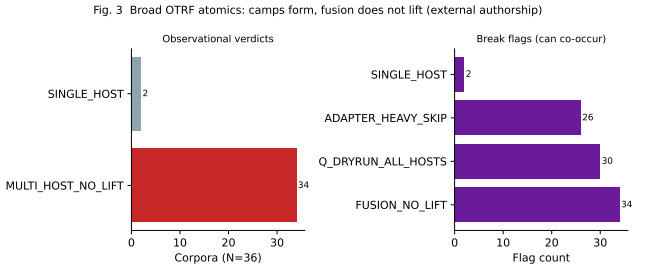


Figure 4. Protocol O (N=36 external). Left: observational verdicts (34 no-lift). Right: co-occurring break flags including 30× fleet-wide quarantine dry-run.

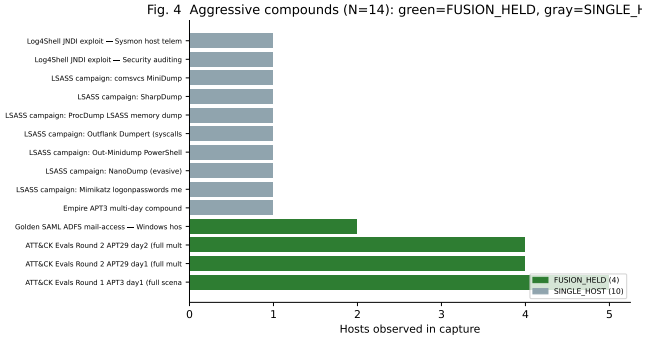


Figure 5. Protocol A (N=14). Green bars: FUSIONHELD (4). Gray: SINGLEHOST (10). x-axis: hosts observed in capture.

large (+0.441), but absolute quality is no longer a hero number.

Do not confuse lift with readiness.. Fig. 2 shows breaktest lift can *exceed* sealed lift while absolute F1 falls. A reader who quotes only lift can still overclaim.

6.3. Protocol O — broad OTRF (external)

Honest reading: multi-host campaigns *form* on public captures; fusion does *not* beat detector-only host coverage on this atomic set (34/36 MULTIHOSTNOLIFT; 2 SINGLEHOST). Break flags co-occur: 34× fusion-no-lift, 30× quarantine dry-run proposing essentially all hosts, 26× heavy adapter skip. Those failures drove the correlator and quarantine fixes measured in Protocol A—not a retroactive rewrite of Fig. 4.

Why this is the independence headline.. Protocol O packs are externally authored. They are still lab purple-team captures, not enterprise estates—but they are not Corvex-written fleet manifests. Leading with Fleet20 would reverse the authorship hierarchy (Section 7).

6.4. Protocol A — aggressive compounds

Held set: APT29 day1, APT29 day2, Caldera APT3 day1, Golden SAML (Table 4). The remaining 10 corpora are SINGLEHOST (Empire APT3 compound, LSASS×7, Log4Shell×2)—fusion not testable. After ambient micro-exfil poison, hop peers via WorkstationName, no recon host-union, and surgical cut-point quarantine, fleet-wide Q dry-run and MULTIHOSTNOLIFT counts on this aggressive set are zero in the locked rollup—but adapter skip rates on APT29 remain 0.86–0.94.

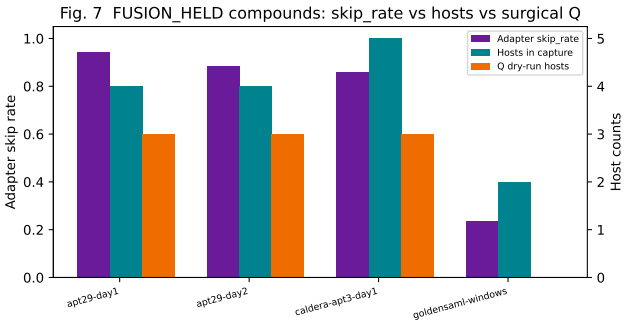


Figure 6. Even held compounds: adapter `skip_rate` (purple) vs hosts (teal) vs surgical Q dry-run host count (orange). APT29 remains allowlist-blind on most raw events.

TABLE 4. PROTOCOL A HELD COMPOUNDS (LOCKED). SKIP RATES FROM ADAPTER LOGS.

Corpus	Hosts	Events	Skip rate	Q hosts
APT29 day1	4	10 980	0.944	3
APT29 day2	4	67 693	0.885	3
Caldera APT3 day1	5	14 423	0.859	3
Golden SAML	2	29	0.237	0

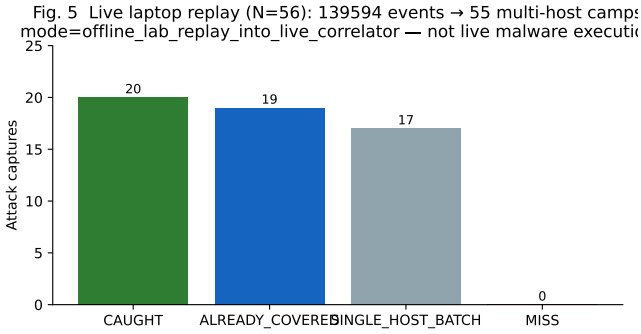


Figure 7. Protocol L catch labels from locked live stress JSON. Mode=`offline_lab_replay_into_live_correlator`. 139,594 events → 55 multi-host camps. Not live malware.

Saturated-lab honesty.. On saturated 4–5 host labs, host-coverage lift can be definitionally zero when both arms already cover all hosts. FUSIONHELD here means consolidation + surgical Q (not fleet-wide spray), not "more hosts than detector on every atomic."

6.5. Protocol L — live laptop replay

Wall time for the locked run is 675 s. Final snapshot: 139,594 events; 55 multi-host campaigns spanning APT29, Caldera, Mordor workstation, and Golden SAML host-

TABLE 5. PROTOCOL L CATCH ROLLUP (N=56).

Label	N	Meaning
CAUGHT	20	New multi-host bridge / camp growth
ALREADYCOVERED	19	Hosts already in multi-host camps
SINGLEHOSTBATCH	17	Capture has < 2 hosts
MISS	**0**	Multi-host capture failed to register

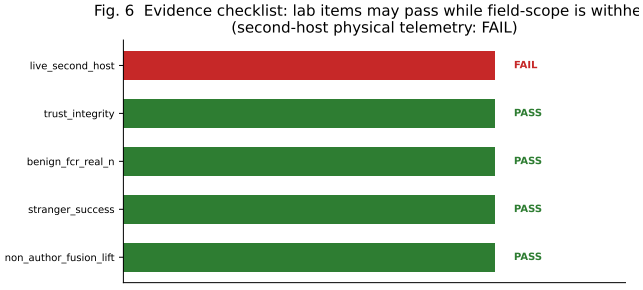


Figure 8. Evidence checklist (locked reports). Four items pass; second-host physical telemetry fails. Lab scores may be green while field-scope prose remains unwarranted.

TABLE 6. MECHANISM TABLE (SYNTHETIC UNLESS NOTED). STATUS AS OF LOCKED REPORTS.

Mode	Mechanism	Status
DHCP split-brain	Distinct `host_id` strings	Held
Day-scale sleeper	1 h resume fragmented chains	Held
Shared svc glue	Time window glued hop islands	Unit held
Egress bury	CDN-like dst fanout	Mixed / Broke cells
DNS-only C2	No Sysmon EID 22	Honest gap
Fleet-wide Q	Recon union + all-host spray	Fixed
Ambient metadata	`169.254.169.254`/Azure wire	Fixed
APT29 skip	Channel allowlist	Remaining

names. ALREADYCOVERED is not failure: later captures land inside camps grown by earlier ones. MISS $= 0$ must not be read as "caught live malware on the laptop."

6.6. Named failure mechanisms

Earlier packaging named breaks without mechanisms. Table 6 is the condensed WHY list. Egress bury remains a live edge on some intensive author cells even after unit poison tests pass—another reason not to lead with author-fleet headlines.

7. Secondary Author Fleets (Not Independence)

Author-written fleets are useful for mechanism debugging. They are the wrong headline for independence.

Fleet20.. Locked rollup: $20/20$ HELD, fragile rate 0.05 among held cells. Never report bare $20/20$ without the fragile rate.

Intensive multifactor.. Locked rollup: HELD 23 (fragile 5), PARTIAL 2, BROKE 3 (all `exfil_bury` cells), errors 0. These packs are in-repo purple-team sketches under `labs/breaktest/manifests/fleet-intensive`.

Placement rule.. If an abstract mentions Fleet20 before OTRF, authorship hierarchy is inverted. This paper places F_{sec} after Protocols O/A/L by design.

8. Discussion: Withheld Field-Scope Sentence

Example of an unwarranted upgrade.. "Corvex works on real multi-host attacks in enterprise networks and is ready for production SOC use." Under the locked evidence checklist, second-host physical telemetry is missing, so that sentence is out of scope for this manuscript. The standing scope remains synthetic ATT&CK-shaped fleets and externally authored lab purple-team captures.

What lab-coherence buys.. Sealed F1 1.00, breaktest lift $+0.441$, stranger dry-run attestation, and integrity probes support statements of the form: *under these protocols, fusion beats detector-only and ingest rejects bad HMAC*. They do not buy enterprise readiness.

Reusable checklist (for other detectors)..

1) Authorship column on every scoreboard.
2) Micro-F1 only for headlines ($2PR/(P+R)$ from pooled counts).
3) Always ablate fusion (or the analogous upgrade).
4) External corpora before self-fleets.
5) Name failure mechanisms—never status labels alone.
6) Separate lab-coherent scores from field-scope warrant.
7) Quarantine cut-points (fleet-wide dry-run is not ready).
8) Replay honesty (replay $\neq$ live implant catch).

Implications for detection papers.. Negative or partial results on external corpora are not defects to bury under sealed F1. They are the content that makes the sealed number interpretable.

9. Related Work

9.1. Closed-world evaluation

Sommer and Paxson warn that closed-world intrusion evaluations overstate detector quality [1]. The analogy here is *sealed-pack optimism*: a separable held-out pack proves wiring and is then screenshotted as field evidence.

9.2. Datasets and purple-team corpora

Surveys of network intrusion datasets emphasize authorship, labeling, and leakage risks [4]. OTRF/Security-Datasets and ATT&CK Evaluations provide external purple-team captures used here as independence stress [6]–[8]. Atomic Red Team supplies public TTP IDs referenced by breaktest manifests [5].

9.3. Reporting discipline

Model cards and explicit release criteria motivate separating score tables from field-scope prose [9]. CSET-style venues solicit metrics, datasets, and experimental discipline [3]. We package these practices into one open correlator evaluation with locked figures—not a novel learning algorithm.

9.4. Negative results

Reliability and security tooling have structural incentives to publish hero lab scores and soft-pedal external drops. We write against that incentive by leading with Protocol O/A and by withholding field-scope sentences when second-host evidence is absent.

10. Threats to Validity

Internal.. Sealed and breaktest ground truth are author-defined; micro-F1 inherits those labels. Observational OTRF verdicts avoid inventing GT F1 but depend on adapter allowlists and host-coverage definitions.

External.. OTRF and ATT&CK Evals captures are lab purple-team, not enterprise estates. Live replay is not live malware execution. Second-host physical telemetry is missing.

Construct.. FUSIONHELD on saturated fleets is not identical to positive host-coverage lift. Catch labels on Protocol L mix growth and already-covered states; both are reported.

Conclusion.. The paper's primary claim is about evaluation hygiene and scope boundaries, which is less sensitive to enterprise transfer than a "beats commercial EDR" claim would be—a claim we do not make.

11. Limitations

Sealed N_{atk}=6 is modest; larger N reduces variance but does not create field validity. Breaktest remains author-assembled. OTRF is external but still lab purple-team. FUSIONHELD $\neq$ host-coverage lift on every atomic. Adapter skip rates on APT29 remain high. No second physical Windows host yet—field-scope sentences withheld. Home-lab benign PASS/FAIL against pre-committed bars deferred. Live stress is replay. Author fleets remain secondary.

12. Conclusion

Sealed F1 1.00 is not a field claim. Under locked protocols, Corvex shows a measurable drop on breaktest, honest no-lift on broad OTRF, partial holds on hardest compounds with residual adapter blindness, and zero-miss live *replay*, while second-host evidence remains absent. The portable lesson is the protocol ladder and authorship hierarchy: seal, ablate, prefer external corpora, name mechanisms, and withhold field-scope prose until the stated evidence exists. Future work: second-host attestation and benign baseline PASS/FAIL—without softening scope ahead of evidence.

Artifact Availability

https://github.com/Siddarthb07/corvex; figures: `docs/paper/figures/`; lockfile: `docs/paper/NUMBERS_LOCKED.json`; rebuild figures: `python scripts/render_paper_figures.py`; rebuild PDF: `python scripts/build_journal_paper_pdf.py`.

Ethics

No live malware execution; OTRF sanitized before local storage; quarantine dry-run by default; TLP:CLEAR public purple-team data [10].

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Appendix Reproduce

```
pip install -e ".[dev]"
corvex seal-day0 --force && corvex eval --split heldout
corvex score-non-author
python scripts/fetch_opensource_corpora.py --full
python scripts/run_opensource_attack_eval.py
python scripts/fetch_opensource_corpora.py --aggressive
python scripts/run_opensource_attack_eval.py --aggressive
python scripts/run_live_opensource_stress.py --dash
python scripts/render_paper_figures.py
```

Appendix Authorship Tiers